Course File IE407 – Internet of Things

Course Title	Internet of Things
Course Code	IE 407
Credit Structure (L-T-P-Cr)	3-0-2-4
Program and Semester	B.Tech 6th Semester
Category	ICT Elective
Instructor	Manish Chaturvedi; Office: FB1-111; Email: manish_chaturvedi@dau.ac.in
Prerequisite courses	IT305 Computer Networks, EL203 Embedded Hardware Design
Content Outline	Internet of Things (IoT) enables embedded smart devices to be an integral part of the Internet. In this course, we study various aspects of IoT system design including hardware and software building blocks, communication protocols, and computation frameworks. The course begins by discussing a variety of IoT enabled applications identifying key issues in designing end to end IoT applications. We discuss the preliminaries of wireless communication and wireless MAC layer protocols. After the basics, the course focuses on the IoT system specific issues such as energy constraints, localization, coverage, topology control, time synchronization, mobility, routing, data aggregation and dissemination, etc. Next we look at the building blocks of IoT applications and their characteristics. A number of domain specific IoT applications and their architecture in vehicular, social, and sensor networks are discussed. Data analysis tools and machine learning techniques are introduced to process the large amount of data produced by IoT applications. The computation models for large scale data processing are also discussed. In the lab, we introduce the programming aspects of IoT applications. All students are expected to complete a course projects that involves designing and building a complete IoT application. A combination of reference books and a set of research papers are used to discuss the fundamentals, seminal findings and new directions in IoT research.
Text / Reference Books	• Sudip Misra, Anandarup Mukherjee, Arijit Roy, Introduction to IoT, Cambridge University Press. • Arshdeep Bahga and Vijay Madisetti, Internet of Things – a hands-on approach, Universities Press. • Waltenegus Dargie and Christian Poellabauer, Fundamentals of Wireless Sensor Networks – theory and practice, Wiley Publisher • C. Siva Ram Murthy and B. S. Manoj, Ad Hoc Wireless Networks: Architectures and Protocols, Pearson Education • Few classic research papers discussed in class

Course Outcome	After studying this course, student will be able to • Understand and analyze basic protocols used in IoT applications. • Identify and address the specific challenges encountered in the IoT application design. • Design IoT applications for different domain and be able to analyze their performance. • Implement fully functional IoT applications on embedded and cloud platforms.
Evaluation Scheme	• In-sem Test(s) 20-25% • Final Exam 40% • Lab/Project 35-40%
Program Outcome	P01, P04, P05, P09

Lecture Plan

S.No.	**Topics**	**Lecture Hrs**
1	**Introduction**	3
1.1	Introduction to Embedded Internet and its components	
1.2	Wireless Communication Basics	
1.3	Survey of WSN and IoT Applications	
2	**Link Layer - Wireless Medium Access Control**	3
2.1	Wireless Medium Access issues	
2.2	MAC protocol survey; Low Power MACs; IEEE 802.15.4	
3	**Node addressing, Routing and Transport Protocols**	3
3.1	Basics of 6LoWPAN and IPv6	
3.2	On Demand, Proactive and Hybrid Protocols	
3.3	Transport protocols for low power and noisy media	
4	**Network Bootstrapping**	9
4.1	Sensor deployment, Node discovery protocols	
4.2	Topology control and self organization	
4.3	Node localization	
4.4	Time synchronization	
4.5	Coverage	
5	**Data Aggregation and Dissemination**	6
5.1	In-network Data aggregation	
5.2	Address and content based dissemination of data	
5.3	Energy efficient data aggregation protocols	

6	**IoT System Design and M2M protocols**	6
6.1	Application Design Templates and Implementations	
6.2	M2M Protocols: MQTT, CoAP, XMPP	
6.3	Edge, fog, and cloud computing for large scale data processing	
6.4	Data Analytics for IoT Applications	
7	**Case studies of IoT application design**	6
7.1	Home automation, Smart Agriculture, Healthcare and activity monitoring, Transportation and logistics	

Lab Experiments:

Basics: IoT cloud programming, experimentation with synthetic data.

Projects: Design of an IoT application. Implementation of the cloud component of the IoT application. Testing and verification with simulated or real world data.

Sample applications: Environmental Monitoring - sound level, COx level, Pollution monitoring of vehicles; Smart transportation – tracking, scheduling, ticketing; Crash detection and prevention; Smart street lightning; Virtual fence for kids; Smart parking- location, timing, challan. smart access; Healthcare and activity tracking; Android front-ends for IoT applications, etc.